

Step 2: long-time b6 threshold is robust to most initial-condition changes

Direct CTL fixed | $b_5 = 1e-4$ | one-at-a-time IC perturbations | $T = 10,000$ days

Main finding: only very low $NK(0)$ noticeably shifts the loose long-time threshold; stricter long-time control threshold is essentially unchanged.

What changed?

Tumor(0), MDSC(0), NK(0),
CTL(0), B(0)
0.2x, 0.5x, 1x, 2x, 5x

One compartment changed at a
time

What was measured?

Threshold for $T(10000) < 1e5$
Threshold for $T(10000) < 1e3$
Near-threshold tumor
trajectories

Why this matters

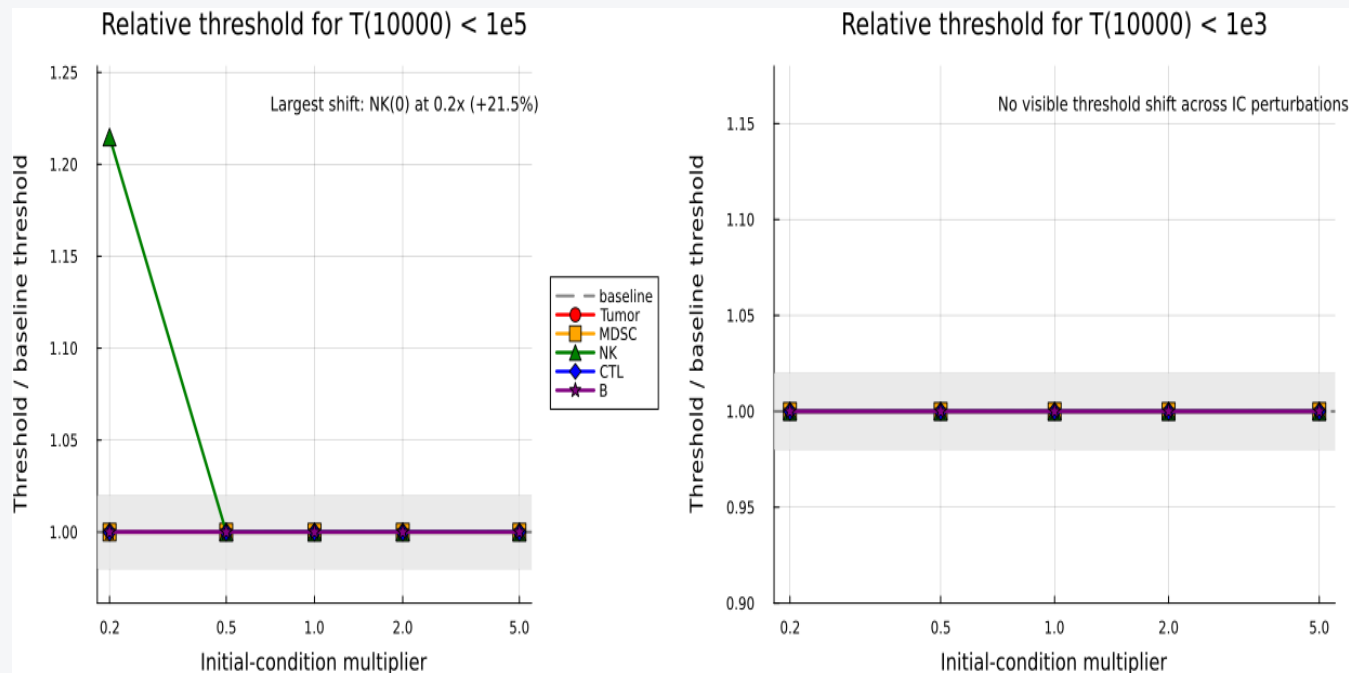
Tests robustness of the b6 switch
Checks if threshold is an IC artifact

IC: initial condition

Q: is the long-time b6 threshold primarily parameter-driven, with $NK(0)$ as the main sensitivity?

Finding 1: most initial-condition changes barely move the long-time b6 threshold

Relative threshold view makes the robustness pattern visible



For $T(10000) < 1e5$, almost all IC perturbations stay near the same b6 threshold.

Only $NK(0) = 0.2x$ baseline pushes the loose threshold upward.

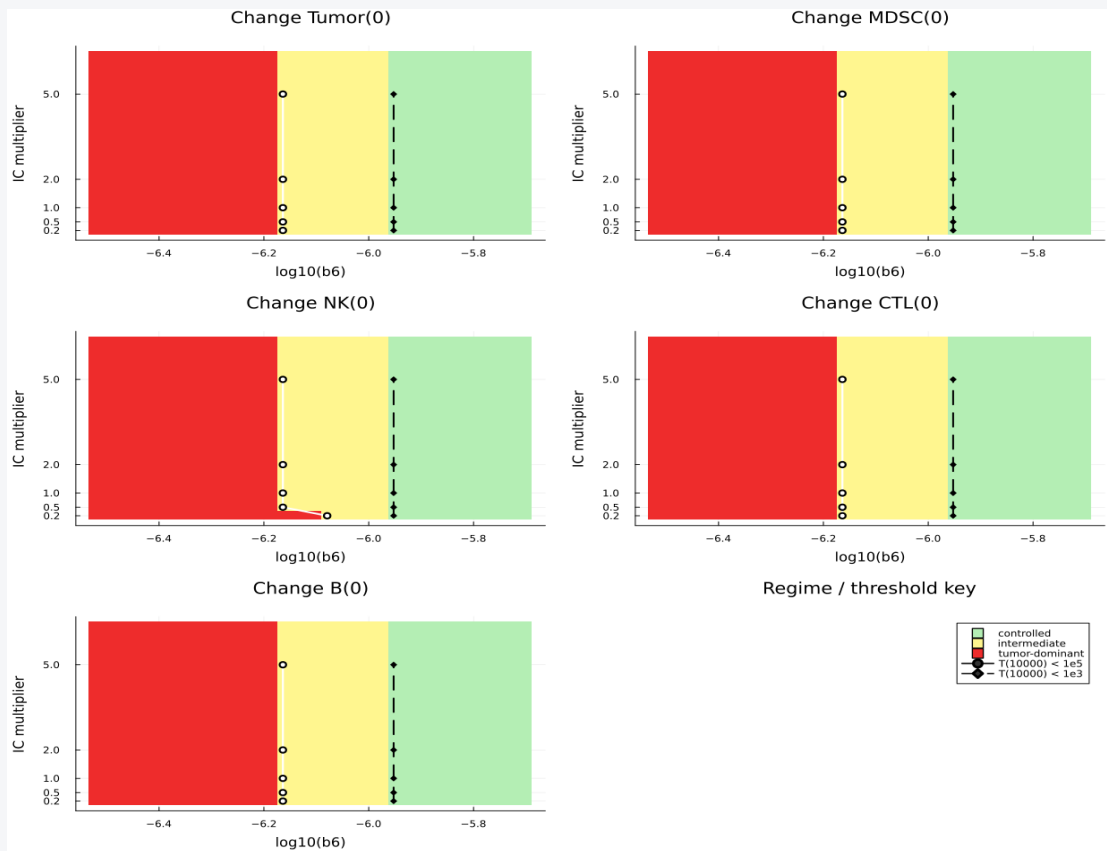
For $T(10000) < 1e3$, the stricter threshold is essentially unchanged across all tested ICs.

Interpretation

- the long-time threshold is robust
- low NK mainly affects how much b6 is needed for partial control.

Finding 2: the long-time phase boundary stays almost vertical across initial-condition changes

Discrete phase map: controlled / intermediate / tumor-dominant regimes



Across Tumor(0), MDSC(0), CTL(0), and B(0), the regime boundary barely shifts.

This means the b_6 switch-like behavior is not just a baseline-IC artifact.

The heatmap is most consistent with a parameter-driven transition plus limited NK sensitivity.

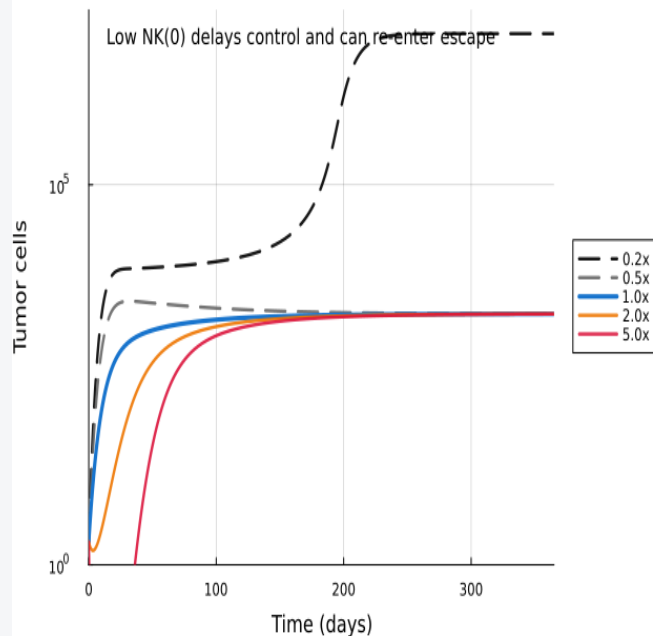
b_6 controls the regime, while most IC changes do not

Finding 3: near the threshold, low NK(0) is the only clearly visible sensitivity

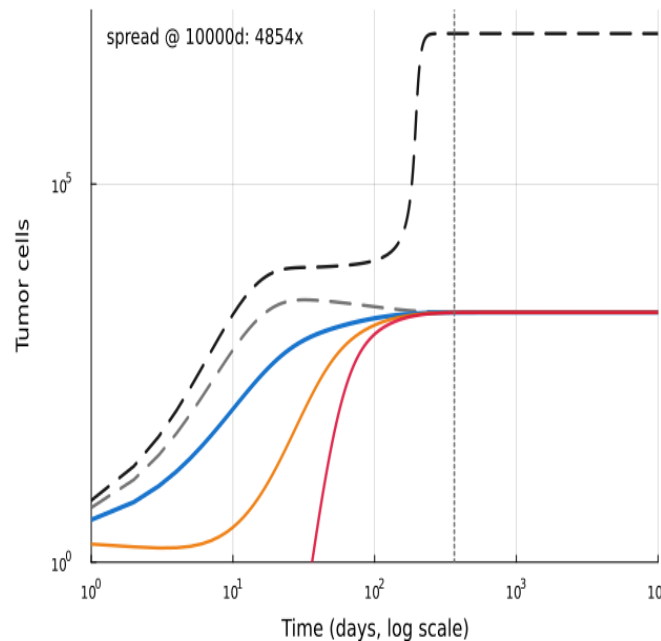
Representative trajectories at $b_6 = 8e-7$

Step 2 highlight: NK(0) is the only IC perturbation with a strong effect near threshold

NK(0) perturbation: early-time divergence



NK(0) perturbation: long-time fate



Varying Tumor(0), MDSC(0), CTL(0), or B(0) gives very similar long-time tumor trajectories

Reducing NK(0) makes tumor control harder near the threshold

NK is the main IC sensitivity to carry forward.

Takeaway: if we extend IC tests, NK-focused perturbations are the most informative.

Summary and next steps

Step 2 confirms robustness

What Step 2 establishes

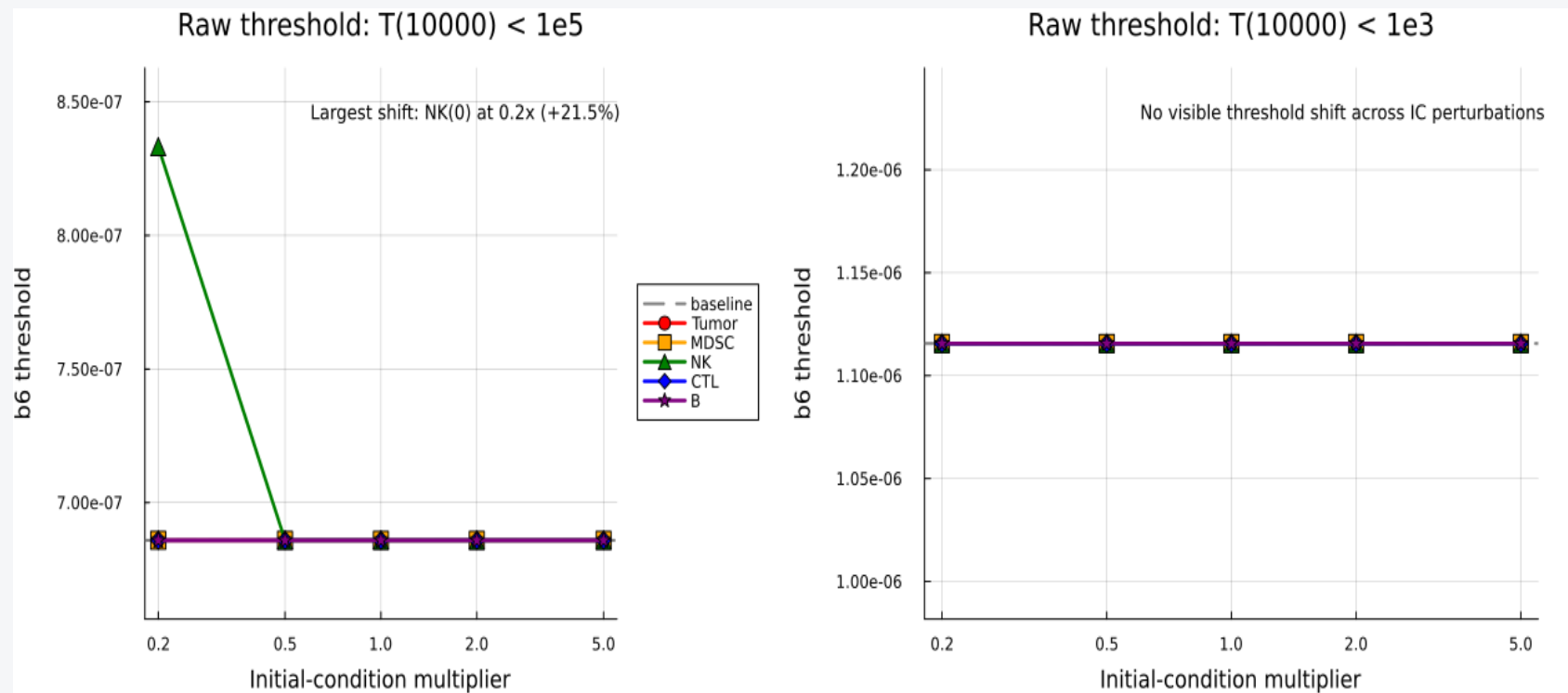
- The long-time b6 threshold persists under most one-at-a-time IC changes
 - The strict control threshold is especially stable
 - Low NK(0) is the only clear sensitivity in the tested range
- > the b6 switch-like behavior is a real model feature, not a baseline-IC artifact

potential next steps

- refine the long-time b5 × b6 phase diagram
- Run up-sweep / down-sweep continuation to test switch vs bifurcation-like behavior
- If more IC testing is needed, prioritize more extreme or combined NK-focused conditions
- Return to SDE only after the long-time ODE transition is fully characterized

Bottom line: keep the direct CTL model, treat b6 as the main parameter, and use NK-focused checks to probe the transition further.

Appendix: raw threshold view



The raw plot is consistent with the relative plot
most thresholds overlap closely, with only low NK(0) showing a noticeable shift